Association between soy and green tea (Camellia sinensis) diminishes hypercholesterolemia and increases total plasma antioxidant potential in dysli...

OBJECTIVE: To evaluate the hypolipemic and antioxidant effects of soy and green tea alone and/or in association in dyslipidemic subjects. METHODS: One hundred dyslipidemic individuals were allocated into four groups. The soy group ingested 50 g of soy (kinako) daily, and the green tea group ingested 3 g of green tea in 500 mL of water per day. A third group ingested 50 g of soy and 3 g of green tea daily, and the control group had a hypocholesterolemic diet. Evaluations were performed at baseline and after 45 and 90 d. Plasma levels of total cholesterol, high-density lipoprotein, and triacylglycerols were evaluated by automated methods. Low-density lipoprotein (LDL) cholesterol was calculated using the Friedewald equation. LDL was isolated by ultracentrifugation. Total plasma antioxidant capacity and plasma levels of total lipid hydroperoxides and those linked to LDL were evaluated by chemiluminescence. The results were expressed as median values and their 25th to 75th percentiles, with a 5% level of significance. RESULTS: No significant difference occurred in LDL, high-density lipoprotein cholesterol, and triacylglycerol levels across groups. However, a statistically significant difference in total cholesterol occurred within the soy/green tea group 45 and 90 d after intervention. No statistically significant difference occurred in plasma levels of lipid hydroperoxides or those linked to LDL in any of the groups studied. All the groups that used soy and/or green tea presented increased total plasma antioxidant potential. CONCLUSION: Soy and green tea, alone or in combination, increased the total antioxidant potential of hypercholesterolemic patients, whereas only the combination decreased total cholesterol levels.